

MMA: Multi-Modal Adapter for Vision-Language Models

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Abstract

Pre-trained Vision-Language Models (VLMs) have served as excellent foundation models for transfer learning in diverse downstream tasks. However, tuning VLMs for few-shot generalization tasks faces a discrimination — generalization dilemma, i.e., general knowledge should be preserved and task-specific knowledge should be fine-tuned. How to precisely identify these two types of representations remains a challenge. In this paper, we propose a Multi-Modal Adapter (MMA) for VLMs to improve the alignment between representations from text and vision branches. MMA aggregates features from different branches into a shared feature space so that gradients can be communicated across branches. To determine how to incorporate MMA, we systematically analyze the discriminability and generalizability of features across diverse datasets in both the vision and language branches, and find that (1) higher layers contain discriminable dataset-specific knowledge, while lower layers contain more generalizable knowledge, and (2) language features are more discriminable than visual features, and there are large semantic gaps between the features of the two modalities, especially in the lower layers. Therefore, we only incorporate MMA to a few higher layers of transformers to achieve an optimal balance between discrimination and generalization. We evaluate the effectiveness of our approach on three tasks: generalization to novel classes, novel target datasets, and domain generalization. Compared to many state-of-the-art methods, our MMA achieves leading performance in all evaluations. Code is at <https://github.com/ZjjConan/Multi-Modal-Adapter>

1. Introduction

Deep networks trained on large-scale datasets [10, 81] have greatly boosted the performance on many vision tasks, such as image classification [13, 19, 27, 37, 56, 64, 66], object detection [17, 24, 53, 54, 83], semantic segmentation [6, 42, 63, 79], and person re-identification [59, 69, 75, 76].

Vision-Language Models (VLMs) [1, 28, 30, 50, 68, 70, 71, 74] have recently been introduced as a class of foundation models. They adopt a holistic approach by jointly processing visual and textual information, thereby fostering a shared understanding of the complex interplay between images and language. In order to establish a cohesive representation space, where positive pairs (i.e., related images and texts) are brought together and negative pairs (i.e., unrelated instances) are separated, VLMs are often trained from extensive web-scale datasets, e.g., 400 million image-text pairs used in Contrastive Language-Image Pretraining (CLIP) [50]. After training on such large-scale data, VLMs often show good generalization ability across diverse downstream tasks without task-specific tuning.

Despite their effectiveness, the massive number of parameters of VLMs makes it difficult to fine-tune them for downstream tasks, especially when only a few data are available in the target domains (i.e., few-shot generalization settings). To effectively adapt these pre-trained VLMs, prompt engineering [50] has become pivotal. Prompt engineering refers to the process of crafting input queries to guide models toward desired outputs. For example, in CLIP [50], a collection of handcrafted text prompts, such as “a photo of a <category>” or “a bad photo of a <category>”, are input for the text encoder to compute category-wise embeddings. Then, these embeddings are matched with the visual embeddings encoded by the image encoder to predict the output class. However, designing good prompts requires rich expert knowledge and enormous time. To circumvent this, many researchers add learnable prompts either into the text [4, 67, 84, 85] or the image encoder [51, 61], or both of them [33]. During model training, only the prompts are optimized, while the whole pre-trained VLMs are frozen. Therefore, prompt learning has gained prominence as it offers a more practical approach to tailor pre-trained VLMs for different downstream tasks.

Besides prompt learning, an alternative approach is to construct lightweight modules, called as adapters, to adapt large-scale pre-trained models [5, 7, 8, 16, 25, 26, 45, 58] to different downstream tasks. In contrast to prompt learning, adapters are shallow networks that enhance model gener-

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alizability via some types of feature fusion. For example, two recent methods – Clip-Adapter [16] and AdaptFormer [7], fuse features by adding the outputs from pre-trained models and the added adapters. Similar to prompt learning, only these extra adapters are optimized during the training phase to avoid overfitting. In addition, unlike prompt learning [39, 40, 84, 85], adapters operate independently of network architectures and allow easy integration into diverse networks, including ResNets [19], ViTs [13], Swins [41], Diffusion Models [45] and so on.

Although these adapters are an efficient tool in many NLP and vision applications, they still have two limitations. First, most of existing adapters, such as LoRA [26] and AdaptFormer [7] are based on uni-modal information. For VLMs such as CLIP [50], dual-modal signals – vision and language – coexist and jointly contribute to final predictions. A simple approach is to apply an adapter independently to both modalities. However, this approach does not consider the relationship between text and image representations (before the final predictions). Therefore, direct applications of the same adapters may be insufficient to learn task-specific cues that vary across both vision and language. Second, existing methods do not consider the characteristics of text and image representations. Transfer learning in general faces the discrimination and generalization dilemma — that is, fine-tuning the features that are discriminable across tasks and preserving the features that are general across tasks. For example, AdaptFormer [7] incorporates adapters to every transformer block. This approach works well with a sufficient amount of training data, but may suffer from overfitting problem when training data is scarce. As such, it is important to consider the characteristics of features (*i.e.*, discriminability and generalizability).

To this end, we propose a novel **Multi-Modal Adapter (MMA)** architecture for VLMs such that the text and image representations can be better aligned. Our MMA contains independent projection layers to learn task-specific knowledge in both text and vision branches. To promote different modal alignment, we design a unified feature-projection layer shared by both modalities. During the fine-tuning, this unified feature space communicates gradients from both modalities to improve the alignment. In addition, we evaluate the discriminability and generalizability of features in both branches across datasets. This dataset-level feature classification identifies that higher layer features as task-specific features should be fine-tuned, while lower layer features as pre-trained generalizable features should be frozen. Therefore, we only incorporate our adapters to higher layers. This design circumvents the issue of insufficient statistical information to understand feature characteristics due to limited number of training samples in each dataset. In summary, the main contributions are:

- We introduce a dataset-level analysis method to system-

atically examine feature representations for transformer-based CLIP models. This analysis helps build more effective and efficient adapters for VLMs.

- We propose a novel adapter that contains separate projection layers to improve feature representations for image and text encoders independently. We also introduce a shared projection to provide better alignment between vision-language representations.
- We integrate our adapter into the well-known CLIP model and evaluate them on various few-shot generalization tasks. Experiment results show that our method achieves leading performance among all compared approaches.

2. Related Work

Vision-Language Models. Recent advancements in VLMs have significantly impacted the field of computer vision, particularly in tasks that combine language with images. Representative models include but not limited to CLIP [50], ALIGN [30], FILIP [68], Florence [71], LiT [74], and Kosmos [28, 49]. These models leverage self-supervised paradigm from massive web-scale multi-modal data for training. For example, CLIP [50] and ALIGN [30] are trained with contrastive loss [47] from approximately 400 million and 1 billion image-text pairs, respectively. By collecting more number of multi-modal data [55], these models show promising performance in various downstream applications [29]. Despite their ability to learn generalized representations, efficiently adapting these pre-trained VLMs for specific downstream tasks remains a significant challenge, especially in few-shot settings. To do so, numerous studies have been proposed for different tasks such as few-shot image recognition [16, 35, 77, 84], object detection [15, 18, 72, 80, 86], and segmentation [12, 20, 82]. In contrast, this work proposes a new multi-modal adapters to effectively adapt VLMs in the few-shot generalization tasks.

Efficient Transfer Learning for VLMs. To transfer pre-trained models to downstream tasks, conventional methods [3, 11, 19] fine-tune all parameters of the pre-trained networks. However, as model sizes expanding, the traditional paradigm is inevitably constrained by the substantial computational burden. Moreover, fine-tuning such massive number of trainable parameters has introduced the risk of severe over-fitting, especially in the few-shot settings. Therefore, multiple parameter efficient methods have been introduced in NLP community [25, 26, 40], which are further extended in vision [7, 31] and VLMs communities [16, 33, 38, 73, 84, 85]. These works could be mainly categorized into the token-based prompt learning and network-based adapting (adapter). For prompt learning in VLMs, it initially involves providing textual instructions to the language component of the VLMs. This approach enhances the model’s task comprehension and adaptability. For ex-

ample, CoOp [84] improve the CLIP’s model in few-shot learning by optimizing a continuous set of prompt vectors in its language branch. CoCoOp [85] further extends CoOp by conditioning the prompts on specific image instances. Other representative works allow to capture the distribution of diverse prompts [43], reduce the risk of overfitting problem by using the pre-trained CLIP as general knowledge to regularize learning process [4, 34, 67], and construct multi-modal prompts on both image and text branches [33]. These works significantly improve alignment between vision and language representations, outperforming CoOp and CoCoOp in various aspects. For adapters, existing works often utilize uni-modal adapter for tuning. For example, Clip-Adapter [16] and Tip-Adapter [78] add an adapter layer after the image encoder. Recently, a multi-modal adapter [32] has been introduced for text-video retrieval. It is inserted after every self-attention and feed-forward MLP modules. All these developments for VLMs signify a paradigm shift from full fine-tuning [50] to partially learning-based methods. However, these methods underscore the different behaviours of different layers in leveraging the full potential of large pre-trained VLMs for diverse and challenging downstream tasks, especially for few-shot generalization tasks.

3. Methods

Following most of existing studies [4, 33, 34, 38, 84, 85], we base on the pre-trained transformer-based CLIP models [50], *i.e.*, using transformers in both text and vision encoders. In the following, we first introduce preliminary knowledge on CLIP and then present our proposed MMA.

3.1. Preliminary

CLIP [50] is a fundamental Vision-Language Model (VLM) that has attracted considerable attention in natural language processing and computer vision. It consists of a text branch with an encoder $\mathcal{T}$ and a vision branch with an encoder $\mathcal{V}$. The two branches allow it to understand and bridge the semantic gap between textual descriptions and visual contents. The text and vision encoders are jointly pre-trained with contrastive objective [47, 50] on web-scale image-text pairs [50] to pull related image-text pairs closely, and vice versa for unrelated pairs. By this large-scale pre-training, CLIP can simultaneously encode images and text descriptions to perform a wide range of downstream tasks. Particularly, an Image I will be fed into the image encoder $\mathcal{V}$ to obtain the image feature x as follows:

$$x_0 = \text{PatchEmbed}(I) \quad (1)$$

$$[c_i, x_i] = \mathcal{V}_i([c_{i-1}, x_{i-1}]) \quad i = 1, 2, \dots, L \quad (2)$$

$$x = \text{PatchProj}(c_L) \quad (3)$$

Here, *PatchEmbed* first splits the input image I into fixed-size patches, and then project these patches into features.

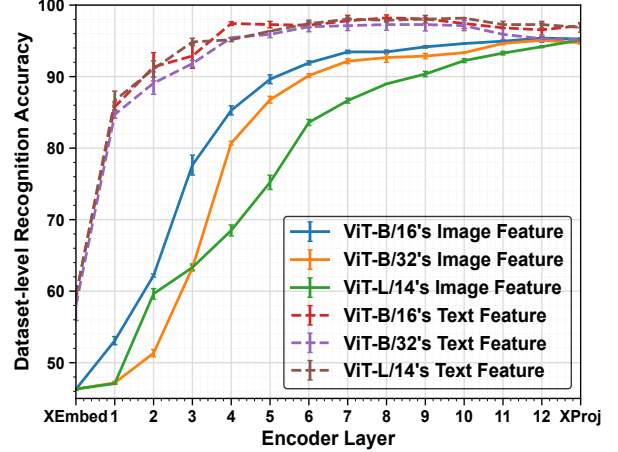


Figure 1. Dataset-level recognition accuracy of different layers in various transformer-based CLIP models. This experiment is to identify which dataset that sample belongs to. We run three times with different seeds, and report average and standard deviation of recognition accuracy for each layer. *XEmbed* refers to text or image embedding layer before the transformer blocks (*i.e.*, self-attention and feed-forward layers [13]), while *XProj* refers to text or image projection layer. Notice that, this experiment only uses training examples from all datasets for evaluation.

After that, a learnable class token c_0 is concatenated with these features – $[c_0, x_0]$, and the concatenated features are sequentially passed through L transformer blocks $\{\mathcal{V}\}_{i=1}^L$. Finally, a projection layer *PatchProj* projects the class token c_L of the last transformer block $\mathcal{V}_L$ into the image feature x , which should lie in the common vision-language space. Similarly, a text description T will be fed into the text encoder $\mathcal{T}$ to obtain text feature w as follows:

$$[w_0]_{j=1}^N = \text{TextEmbed}(T) \quad (4)$$

$$[w_i]_{j=1}^N = \mathcal{T}_i([w_{i-1}]_{j=1}^N) \quad i = 1, 2, \dots, L \quad (5)$$

$$w = \text{TextProj}(w_L^N) \quad (6)$$

As shown, this process has three steps: a *TextEmbed* is used to tokenize and project the input text description into N word embeddings, a series of transformer blocks $\{\mathcal{T}\}_{i=1}^L$ is to abstract features, and a *TextProj* is to project the last token w_L^N of the last transformer block $\mathcal{T}_L$ to the common vision-language space. Given those features, we can compute the cosine similarity scores $\text{sim}(x, w)$ between images and text descriptions in different domains or tasks to perform task-specific predictions.

3.2. MMA: Multi-Modal Adapter

Our work mainly focuses on few-shot generalization tasks [85], where the pre-trained CLIP models are firstly tuned on some base classes with limited training examples, and then directly tested to recognize unseen instances, *e.g.*, novel classes or different types of datasets. For these tasks, it

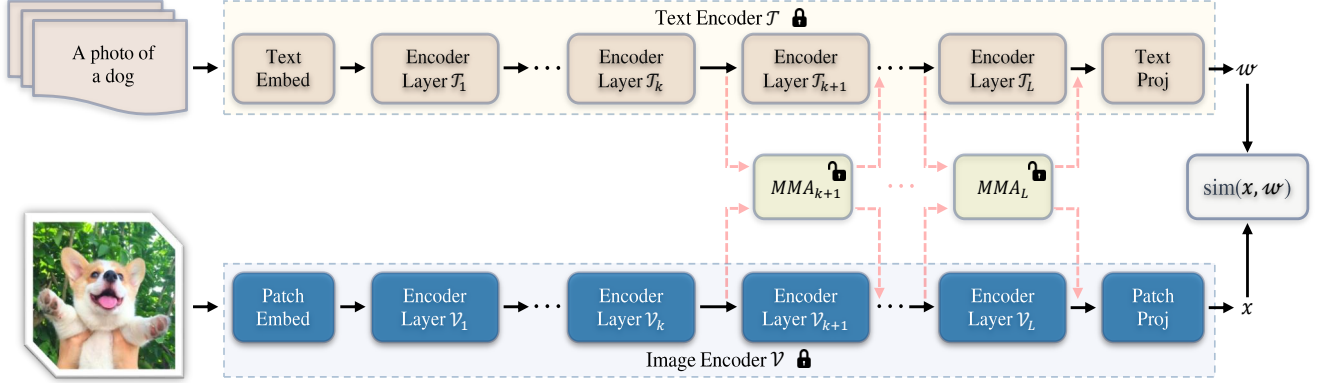


Figure 2. The proposed Multi-Modal Adapter (MMA) for the transformer-based CLIP models. Our MMA tunes both image and text encoders. Only the extra adapters are optimized, while the whole pre-trained CLIP models are frozen. In our method, only a few higher layers ($\geq k$) of each encoder will be tuned based on our analysis to strike a good balance between discrimination and generalization dilemma. Moreover, our MMA shares weights between image and text representations to learn shared cues from different branches. By this design, our MMA eliminates feature-wise interactions between each image-text pair [85], greatly reducing the computational cost.

is well known that a good representation of an instance should be *discriminable*, and also *generalizable* across different types of datasets. These two properties play an important role in transfer learning. Unfortunately, it is difficult to systematically quantify these two characteristics in the few-shot scenario, *i.e.*, a few samples can be accessed in the dataset. Inspired by the dataset bias introduced in [60], we introduce a task in which observers identify which dataset a sample belongs to, called dataset-level recognition. In other words, more discriminable features are easier to distinguish between different datasets, whereas more generalizable features are more invariant across datasets. Based on this intuition, we perform an analysis using three pre-trained transformer-based CLIP models [50], *i.e.*, ViT-B/16, ViT-B/32, and ViT-L/14, due to their superior performance [16, 51, 84, 85]. All models have similar structures in image and text encoders as shown in Eq. (1) to Eq. (6), and the number of transformer blocks L is 12. In addition, to deepen our understanding of different features, we extract features from all layers in both text and image encoders, and train linear classifiers to perform dataset-level recognition. As shown in Fig. 1, we have two observations:

Observation-1. In both pre-trained text and image encoders, higher layers contain discriminable dataset-specific representations, while lower layers contain generalizable representations across different datasets. These results suggest that it is easier to tune higher layers for downstream tasks than lower layers, and that freezing lower layers can preserve more generalizable knowledge than higher layers.

Observation-2. In most cases, text features, as they are encoded with semantic category names, are more discriminable across datasets than visual features. In addition, there are larger gaps between text and image features in lower layers than in higher layers. Therefore, we argue that it is more difficult to align lower layers between text and image

features than between higher layers, especially tuning with limited training samples.

Based on the above two observations, we propose a new adapter-based efficient tuning framework as below.

Macro Design. According to the *observation 1*, we propose a novel Multi-Modal Adapter (MMA) as shown in Fig. 2. Different from most of existing methods that add adapters or tokens to the whole network [7, 25, 26, 32] or some lower layers [33, 34, 84, 85], the new adapter $\mathcal{A}$ (detailed in the next) are partially added into a few higher-layers of both image and text encoders. Formally, for the image encoder $\mathcal{V}$, we add our adapters $\mathcal{A}^v$ from the k -th transformer block and modified Eq. (2) as follows:

$$\begin{aligned} [c_i, x_i] &= \mathcal{V}_i([c_{i-1}, x_{i-1}]) \quad i = 1, 2, \dots, k-1 \quad (7) \\ [c_j, x_j] &= \mathcal{V}_j([c_{j-1}, x_{j-1}]) + \alpha \mathcal{A}_j^v([c_{j-1}, x_{j-1}]) \\ j &= k, k+1, \dots, L. \quad (8) \end{aligned}$$

Here, underline indicates trainable blocks. α is a coefficient to balance between task-specific knowledge and general pre-trained knowledge. Obviously, $\alpha = 0$ degrades to the original transformer block without integrating any extra knowledge. Similarly, we add adapters $\mathcal{A}^t$ to the text encoder $\mathcal{T}$ and modified Eq. (5) as follows:

$$[w_i^j]_{j=1}^N = \mathcal{T}_i([w_{i-1}^j]_{j=1}^N) \quad i = 1, 2, \dots, k-1 \quad (9)$$

$$[w_i^j]_{j=1}^N = \mathcal{T}_j([w_{i-1}^j]_{j=1}^N) + \alpha \mathcal{A}_j^t([w_{i-1}^j]_{j=1}^N) \quad j = k, k+1, \dots, L \quad (10)$$

Micro Design. Currently, our method adapts adapters independently in both image and text branches to learn task-specific knowledge. However, as our *observation-2*, the large semantic gap between vision and language branches will make the model hard to be aligned, especially when

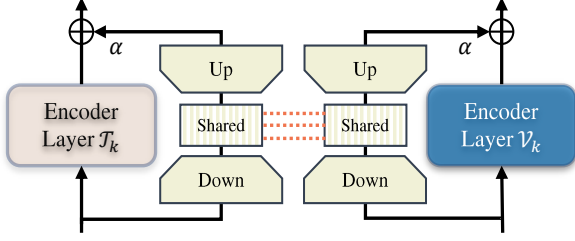


Figure 3. The newly designed multi-modal unit. It contains separate projection layers (“Down” and “Up”) to tune different modals’ encoders, as well as a shared projection layer (“Shared”) to build a strong connection between vision and language branches.

only a few training examples can be accessed. To bridge the representations in both branches, we propose a multi-modal unit with a shared projection layer as shown in Fig. 3. This unit first uses a separate projection layer to project each branch input into features with the same dimensions. After that, a shared projection layer is employed to aggregate these dual-modal signals, followed by a separate layer to match the output dimensions from each branch. Formally, this process can be summarized as follows:

$$\begin{aligned} \mathcal{A}_k^v(z_k) &= \mathbf{W}_{ku}^v \cdot \delta(\mathbf{W}_{ks} \cdot \delta(\mathbf{W}_{kd}^v \cdot z_k)) \\ z_k &= [c_k, \mathbf{x}_k] \end{aligned} \quad (11)$$

A similar process is added to text encoder as follows:

$$\begin{aligned} \mathcal{A}_k^t(z_k) &= \mathbf{W}_{ku}^t \cdot \delta(\mathbf{W}_{ks} \cdot \delta(\mathbf{W}_{kd}^t \cdot z_k)) \\ z_k &= [\mathbf{w}_k^j]_{j=1}^N \end{aligned} \quad (12)$$

Here, $\mathbf{W}_{ku}$ and $\mathbf{W}_{kd}$ are the k -th “Up” and “Down” projection layers illustrated in Fig. 3, where the modality branch is highlighted by superscript. $\mathbf{W}_{ks}$ are the k -th projection layer, which is shared between different branches in Eq. (11) and Eq. (12). Importantly, the shared projection acts as a bridge between two modalities and allows gradients to be propagated into each other, leading to better aligning different modality signals.

4. Experiments

We evaluate the performance of MMA based on previous works [33, 85], including *Generalization from Base-to-Novel Classes*, *Cross-dataset Evaluation*, and *Domain Generalization*. All these experiments are based on 16-shot settings, *i.e.*, only 16 training examples per category.

Generalization from Base-to-Novel Classes. As done in many previous studies [33, 84, 85], we evaluate our method on 11 image classification datasets, including 2 general object recognition datasets: ImageNet [10] and Caltech101 [14]; 5 fine-grained image recognition datasets: OxfordPets [48], StanfordCars [36], Flowers102 [46], Food101 [2], and FGVC Aircraft [44]; scene understanding dataset: SUN397

[65]; a texture dataset: DTD [9]; a satellite-image recognition dataset: EuroSAT [21] and an action classification dataset: UCF101 [57]. These datasets cover a wide range of recognition tasks, which can show good generalization ability of a model. For this experiment, we follow the same setup in [4, 33, 43, 67, 85] that trains our model only on the base classes in a few-shot setting (16-shots), and test the trained model on both base and novel categories.

Cross-dataset Evaluation. Similar to the Base-to-Novel experiments, we also use the aforementioned 11 datasets for cross-dataset evaluation. As suggested in CoCoOp [85], all models are trained on ImageNet with 1000 categories, each category having 16 training samples. After that, models are directly evaluated on other datasets without further tuning.

Domain Generalization. To evaluate the robustness of models on out-of-distribution datasets, Zhou *et al.* [85] suggest to test the ImageNet fine-tuned models on other four variants of ImageNet datasets with different types of domain shifts. These datasets are ImageNetV2 [52], ImageNet-Sketch [62], ImageNet-A [23], and ImageNet-R [22]. We also conduct this experiment for evaluation.

Implementation Details. Following previous works [4, 33, 43, 67, 84, 85], we conduct all experiments with the few-shot setting, *i.e.* 16 shots per category. We use ViT-B/16 based CLIP model in all settings of experiments. In the Base-to-Novel setting, we add the proposed multi-modal unit starting from $k = 5$ transformer block to the last one in both language and vision branches. The dimension of the shared projection layer is 32. We also use the template “a photo of a <category>” [33, 85] for the word embeddings, where “<category>” will be replaced with the class names as zero-shot recognition [50]. We train our models for 5 epochs. On the large-scale ImageNet dataset, we use a batch size of 128 for training. On the other 10 datasets, we set the batch size to 16. For the other two experiment settings, similar to MaPLe [33], we set $k = 9$ and train our models just for 1 epoch. Optimization is done by a SGD solver with a momentum of 0.9 and a weight decay of 0.0005. All our models are trained with a cosine learning rate schedule on a single GPU device with mix-precision for speeding up. We report *Base* and *Novel* class accuracies, and their harmonic mean (*HM*) averaged over 3 runs with 3 different seeds. For other two settings, we report class accuracy on each dataset.

4.1. Main Results

Base-To-Novel Generalization. In this experiment, we compare our MMA with many state-of-the-art approaches, including the zero-shot baseline – CLIP [50], text-based prompt learners – CoOp [84], CoOpOp [85], ProDA [43], KgCoOp [67], LASP [4] and LASP-V [4], and two recently introduced multi-modal prompt learning methods: RPO [38] and MaPLe [33]. Recognition accuracy on 11 widely

Table 1. **Comparison with state-of-the-art methods on different datasets in the Base-to-Novel Generalization setting.** “Base” and “Novel” are the recognition accuracies on base and novel classes respectively. “HM” is the harmonic mean of base and new accuracy, providing the trade-off between adaption and generalization. The proposed MMA shows a good adaptation ability, while being highly effective in novel class generalization. The entries noted by grey are obtained by using novel class information during training.

Methods	Average			ImageNet			Caltech101			OxfordPets		
	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM
CLIP [ICML2021] [50]	69.34	74.22	71.70	72.43	68.14	70.22	96.84	94.00	95.40	91.17	97.26	94.12
CoOp [IJCV2022] [84]	82.69	63.22	71.66	76.47	67.88	71.92	98.00	89.81	93.73	93.67	95.29	94.47
CoOpOp [CVPR2022] [85]	80.47	71.69	75.83	75.98	70.43	73.10	97.96	93.81	95.84	95.20	97.69	96.43
ProDA [CVPR2022] [43]	81.56	72.30	76.65	75.40	70.23	72.72	98.27	93.23	95.68	95.43	97.83	96.62
KgCoOp [CVPR2023] [67]	80.73	73.60	77.00	75.83	69.96	72.78	97.72	94.39	96.03	94.65	97.76	96.18
MaPLe [CVPR2023] [33]	82.28	75.14	78.55	76.66	70.54	73.47	97.74	94.36	96.02	95.43	97.76	96.58
LASP [CVPR2023] [4]	82.70	74.90	78.61	76.20	70.95	73.48	98.10	94.24	96.16	95.90	97.93	96.90
LASP-V [CVPR2023] [4]	83.18	76.11	79.48	76.25	71.17	73.62	98.17	94.33	96.43	95.73	97.87	96.79
RPO [ICCV2023] [38]	81.13	75.00	77.78	76.60	71.57	74.00	97.97	94.37	96.03	94.63	97.50	96.05
MMA [this work]	83.20	76.80	79.87	77.31	71.00	74.02	98.40	94.00	96.15	95.40	98.07	96.72

Methods	StanfordCars			Flowers102			Food101			FGVCAircraft		
	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM
CLIP [ICML2021] [50]	63.37	74.89	68.65	72.08	77.80	74.83	90.10	91.22	90.66	27.19	36.29	31.09
CoOp [IJCV2022] [84]	78.12	60.40	68.13	97.60	59.67	74.06	88.33	82.26	85.19	40.44	22.30	28.75
CoOpOp [CVPR2022] [85]	70.49	73.59	72.01	94.87	71.75	81.71	90.70	91.29	90.99	33.41	23.71	27.74
ProDA [CVPR2022] [43]	74.70	71.20	72.91	97.70	68.68	80.66	90.30	88.57	89.43	36.90	34.13	35.46
KgCoOp [CVPR2022] [67]	71.76	75.04	73.36	95.00	74.73	83.65	90.50	91.70	91.09	36.21	33.55	34.83
MaPLe [CVPR2022] [33]	72.94	74.00	73.47	95.92	72.46	82.56	90.71	92.05	91.38	37.44	35.61	36.50
LASP [CVPR2022] [4]	75.17	71.60	73.34	97.00	74.00	83.95	91.20	91.70	91.44	34.53	30.57	32.43
LASP-V [CVPR2022] [4]	75.23	71.77	73.46	97.17	73.53	83.71	91.20	91.90	91.54	38.05	33.20	35.46
RPO [ICCV2023] [38]	73.87	75.53	74.69	94.13	76.67	84.50	90.33	90.83	90.58	37.33	34.20	35.70
MMA [this work]	78.50	73.10	75.70	97.77	75.93	85.48	90.13	91.30	90.71	40.57	36.33	38.33

Methods	SUN397			DTD			EuroSAT			UCF101		
	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM
CLIP [ICML2021] [50]	69.36	75.35	72.23	53.24	59.90	56.37	56.48	64.05	60.03	70.53	77.50	73.85
CoOp [IJCV2022] [84]	80.60	65.89	72.51	79.44	41.18	54.24	92.19	54.74	68.69	84.69	56.05	67.46
CoOpOp [CVPR2022] [85]	79.74	76.86	78.27	77.01	56.00	64.85	87.49	60.04	71.21	82.33	73.45	77.64
ProDA [CVPR2022] [43]	78.67	76.93	77.79	80.67	56.48	66.44	83.90	66.00	73.88	85.23	71.97	78.04
KgCoOp [CVPR2023] [67]	80.29	76.53	78.36	77.55	54.99	64.35	85.64	64.34	73.48	82.89	76.67	79.65
MaPLe [CVPR2023] [33]	80.82	78.70	79.75	80.36	59.18	68.16	94.07	73.23	82.35	83.00	78.66	80.77
LASP [CVPR2023] [4]	80.70	78.60	79.63	81.40	58.60	68.14	94.60	77.78	85.36	84.77	78.03	81.26
LASP-V [CVPR2023] [4]	80.70	79.30	80.00	81.10	62.57	70.64	95.00	83.37	88.86	85.53	78.20	81.70
RPO [ICCV2023] [38]	80.60	77.80	79.18	76.70	62.13	68.61	86.63	68.97	76.79	83.67	75.43	79.34
MMA [this work]	82.27	78.57	80.38	83.20	65.63	73.38	85.46	82.34	83.87	86.23	80.03	82.20

used datasets of base (**Base**) and novel (**Novel**) classes, as well as the trade-off between these two metrics – harmonic mean (HM), are reported in Tab. 1.

Based on these results, we have made three main conclusions. First, the proposed MMA achieves the best average performance over 11 datasets on all evaluation met-

rics, *i.e.*, base and novel accuracy, as well as their harmonic mean. Among all compared methods, LASP [4] introduces a text-to-text loss that maximizes the probability of the learned prompts to be correctly classified into pre-defined handcrafted textual prompts. This method provides the best trade-off in recognizing base and novel classes,

Table 2. **Comparison of MMA with state-of-the-art methods in the Cross-Dataset Evaluation setting.** Overall, our MMA obtains leading average performance over 10 datasets, demonstrating the good zero-shot transferable ability.

Methods	ImageNet	Caltech101	OxfordPets	StanfordCars	Flowers101	Food101	FGVCAircraft	SUN397	DTD	EuroSAT	UCF101	Average
CoOp [IJCV2022] [84]	71.51	93.70	89.14	64.51	68.71	85.30	18.47	64.15	41.92	46.39	66.55	63.88
CoCoOp [CVPR2022] [85]	71.02	94.43	90.14	65.32	71.88	86.06	22.94	67.36	45.73	45.37	68.21	65.74
MaPLe [CVPR2023] [33]	70.72	93.53	90.49	65.57	72.23	86.20	24.74	67.01	46.49	48.06	68.69	66.30
PromptSRC [ICCV2023] [34]	71.27	93.60	90.25	65.70	70.25	86.15	23.90	67.10	46.87	45.50	68.75	65.81
MMA [this work]	71.00	93.80	90.30	66.13	72.07	86.12	25.33	68.17	46.57	49.24	68.32	66.61

Table 3. **Comparison of MMA with state-of-the-art methods in the Domain Generalization setting.** Overall, our MMA obtains the best performance in 3/4 out-of-distribution datasets, showing good robustness to domain shifts.

Methods	ImageNet	-V2	-S	-A	-R
CLIP [ICML2021] [50]	66.73	60.83	46.15	47.77	73.96
CoOp [IJCV2022] [84]	71.51	64.20	47.99	49.71	75.21
CoCoOp [CVPR2022] [85]	71.02	64.07	48.75	50.63	76.18
MaPLe [CVPR2023] [33]	70.72	64.07	49.15	50.90	76.98
MMA [this work]	71.00	64.33	49.13	51.12	77.32

which is further improved by training with novel class information in LASP-V. Our MMA outperforms LASP and its variant LASP-V with an average HM of 79.87 without any novel class information during training. Specifically, our MMA achieves slightly better performance on the base class (+0.02) and performs significantly better than LASP (+1.9) and LASP-V (+0.69) on novel class generalization. We believe that our method is more flexible and it is much easier to deploy our method in different unseen scenarios than LASP-V. Second, the recent multi-modal prompt learning method - MaPLe [33] adds learnable prompts from lower to higher layers of both text and image encoders. Furthermore, MaPLe uses a couple function to improve the alignment between text and image features. Compared with MaPLe, our MMA achieves +0.92, +1.66, and +1.32 performance gains in terms of base accuracy, novel accuracy, and their harmonic mean, respectively. Third, none of these methods can obtain the leading performance on all 11 datasets in all three evaluation metrics. For example, our MMA obtains better performance on base classes on 8/11 datasets, while also performing better on novel classes on 4/11 datasets; LASP-V obtains the leading performance on 2/11 and 3/11 datasets on base and novel classes respectively; the zero-shot classifier (denoted as CLIP) achieves superior performance on novel class on Flowers102, and also performs on par with others on Caltech101, OxfordPets, and Food101. These results indicate that the Base-to-Novel generalization is still challenging, and our MMA provides the best trade-off.

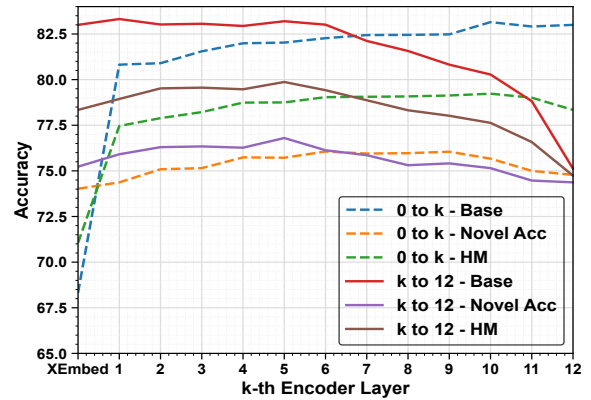


Figure 4. Ablation studies on different choices of adding our proposed multi-modal units. We report average scores of Base, Novel and HM over 11 datasets. Results obtained by adding after $k = 5$ shows the best tradeoff between discrimination and generalization.

Cross-Dataset Evaluation. We also experiment with the cross-dataset setting. Our MMA is first trained on all the 1000 ImageNet categories and then directly evaluated on the other 10 datasets used in the previous experiments. Tab. 2 summarizes all results. Our MMA achieves the best average accuracy of 66.61 compared with other state-of-the-art methods. Particularly, MMA surpasses the second performer - MaPLe in half of these datasets. In addition, On the trained source dataset - ImageNet, our MMA also obtains very competitive performance against CoOp and CoCoOp. MMA is slightly better than MaPLe. These results demonstrate the good zero-shot transferability of our MMA.

Domain Generalization. Following previous works [33, 84, 85], we directly evaluate the models tuned on ImageNet to various out-of-domain datasets and show results in Tab. 2. Our MMA obtains superior performance in 3/4 out-of-distribution datasets, demonstrating the robustness to domain shifts in the domain generalization setting.

4.2. Ablation Experiments

Variants of Adding MMA. We first evaluate the choices to add our MMA to different encoder layers. These choices include two kinds of approaches: from $XEmbed$ to k -th

Table 4. Ablation experiments over 11 datasets used in Base-to-Novel Generalization setting.

(a) Performance with Different Model Variants				(b) Dimensions of Shared Layers				(c) Scaling Factor α			
Model Variants	Base	Novel	HM	Dims	Base	Novel	HM	α	Base	Novel	HM
Only L-Adapter	80.36	75.81	78.02	8	82.66	76.17	79.28	0.0001	79.40	75.57	77.44
Only V-Adapter	80.39	74.18	77.16	16	82.80	76.48	79.52	0.0005	81.81	76.08	78.84
No SharedProj	82.43	76.21	79.20	32	83.20	76.80	79.87	0.001	83.20	76.80	79.87
FCAA [1]	79.11	75.64	77.34	64	83.41	76.17	79.63	0.005	83.80	75.37	79.36
MMA	83.20	76.80	79.87	128	82.98	76.54	79.58	0.01	84.27	74.32	78.98

Table 5. Comparing our MMA with the baseline by fine-tuning last few layers on 11 datasets in Base-to-Novel Generalization setting. “10→12” refers to fine-tune the last 3 layers in both branches.

Layer	12	10→12	8→12	5→12	MMA
Base	80.77	83.02	83.77	83.21	83.20
Novel	74.08	74.55	73.77	70.95	76.80
HM	77.28	78.56	78.45	76.59	79.87

layers, and from k -th layers to the last one, where $k = XEmbed, 1, \dots, 12$. All results are shown in Fig. 4. For the former method, increasing k generally improves the performance on base classes but decreases accuracy on novel classes. For the latter method, the performance on base classes is generally maintained when k is around 5. But there is a large increase in the performance on novel classes before $k = 5$. So the highest HM of is 79.87 at $k = 5$. This further demonstrates our previous findings.

Adapting Variant Options. We assess the efficacy of various design choices for adapters. These design alternatives encompass only uni-modal adapter added in vision (V-) or language (L-), and no shared projection layer (No *SharedProj*). We replace every MMA with Flamingo-style cross-attention [1] to test whether it can be used as efficient adapters (referred as FCAA). In Tab. 4a, we present averaged results across 11 recognition datasets. We find that uni-modal adapters perform worse than the one adding adapters to both branches. Moreover, adding the shared projection layer further increases the HM from 79.20 to 79.87, demonstrating the importance of feature alignment.

Dimension of the Shared Layer. The dimension of shared layers in our MMA determines the number of parameters to extract relationships between the features from the two modalities. We perform an ablation study on MMA by systematically varying the dimensions of the shared layers to investigate its effects. As depicted in Tab. 4b, accuracy on base class is highest with an increment in the middle dimension, but the novel accuracy performance reaches a saturation point at approximately 32. This may be because a larger dimension of the shared layers incurs more trainable parameters, increasing the risk of overfitting.

Scaling Factor α . Scaling factor balances the importance

of the general features and the tasks task-specific features. We systematically assess the effect of the scaling factor, and the results are presented in Tab. 4c. Our MMA attains the best trade-off performance (HM) between base and novel classes with $\alpha = 0.001$. A large scaling factor helps our model to quickly adapt to base classes but shows inferior performance to novel classes, while a smaller scaling factor makes the model hard to tune in downstream tasks.

Fine-tuning last few layers. Lastly, we compare our MMA with the baseline by just fine-tuning last few layers. The results are reported in Tab. 5. Fine-tuning more layers performs good on base classes but show worse performance on novel classes. This is because fine-tuning more layers may impair the general knowledge of the pre-trained VLMs.

5. Conclusion

The adaptation of large-scale VLMs, exemplified by CLIP [50], to downstream tasks presents a formidable challenge, primarily because of the extensive number of trainable parameters juxtaposed with the limited scale of available training samples. In this paper, we propose a Multi-Modal Adapter (MMA) designed for both vision and language branches to enhance alignment between their respective representations. We systematically analyze the discriminability and generalizability of features across datasets in both vision and language branches, because these two characteristics play important roles in transfer learning, especially in the few-shot settings. Based on our analysis, we selectively introduce MMA to specific higher layers of transformers to achieve an optimal balance between discrimination and generalization. We assess the effectiveness of our approach through three representative tasks: generalization to novel classes, adaptation to new target datasets, and unseen domain shifts. Comparisons against other state-of-the-art methods demonstrate that our MMA achieves superior performance across all three types of assessments.

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